

COSH: Closed-Form Onset Search via Hyperbolic Response

Methodology for pre-flight critical-moment identification of a multirotor from ground-contact tip-over excitation

This document specifies the complete identification methodology, from raw signals to the take-off feed-forward moment. Every stage mirrors the implementation in `critical_value_getter_pieewise.py`; the companion analyses (model-fidelity bounds, excitation design, ground-effect bracket) are documented in `fidelity_section.tex` and are only referenced here.

1 Problem setting and notation

A multirotor rests on its landing gear. Ramping the commanded roll (pitch) moment tilts the vehicle until it begins to rotate about the landing-gear contact line — the *pivot*. The applied moment at that *rotation onset* is the critical moment M_{crit} ; comparing the two tip directions yields the CoM offset and the feed-forward moment M_{ff} that compensates take-off. No rig, no motion capture and no stable in-flight data are used.

Symbol	Meaning	Value	Provenance
W, z_{CoM}	weight, CoM height	31.59 N, ~ 0.3 m	measured / conservative
$l_{p,\phi}, l_{p,\theta}$	pivot offsets (roll, pitch)	0.140, 0.110 m	conservative geometry
J_P	inertia about the pivot line	—	never used numerically
C_T	thrust map, $T_i = C_T \Omega_i^2$	1.3175×10^{-7}	bench calibration
M_{thr}	window threshold	0.01 N m	noise floor
M_{cap}	allocator caps (x, y)	2.37, 2.74 N m	allocation feasibility
ε_{max}	ramp-slope gate	3%	sensitivity chain (Sec. 5)
M_{floor}	slope-gate floors (x, y)	0.7, 0.4 N m	$ M_{\text{crit}} $ lower bound, ± 20 mm box
N_{min}	window-sample gate	38	30 pre + 8 post
$e_{\text{lin,max}}$	linearity-RMSE gate	30 mN m	10 $\times$ noise floor
N_{seg}	sweep margin (each end)	8	degenerate-candidate guard

Table 1: Design constants. (C_2, K) are *calibrated*, not set (Sec. 7).

2 Pipeline overview

The complete pipeline is five stages (Fig. 1); the three algorithms nest as Algorithm 3 $\supset$ Algorithm 2 $\supset$ Algorithm 1, and Algorithm 1 is the only stage that touches the angular rate.

1. **Signals and window** (Sec. 4) — per run: reconstruct $M(t)$ from the measured rotor speeds and take $\omega(t)$ from odometry; open the excitation window at $|M| > M_{\text{thr}}$ (backward from the peak), close it at the moment peak, truncated at the allocator cap.
2. **Run gates** (Sec. 5) — discard runs failing the slope-error (3%), sample-count (38) or linearity-RMSE (30 mN m) gates; moment trace only, so no circularity.
3. **Calibration, once per configuration and axis** (Sec. 7) — Algorithm 3 scans (C_2, K) coarse-to-fine; each candidate is scored by Algorithm 2, which runs Algorithm 1 on every gated run and sums the two directional CVs of M_{crit} across ramp rates.
4. **Identification, per run** (Sec. 6) — Algorithm 1 with the calibrated (C_2^*, K^*) pinned, at full sweep resolution: amplitude fixed by $C_1 = K\dot{M}$, baseline by the pre-segment median, the onset index swept exhaustively; $M_{\text{crit}} = M(t_{j^*})$.

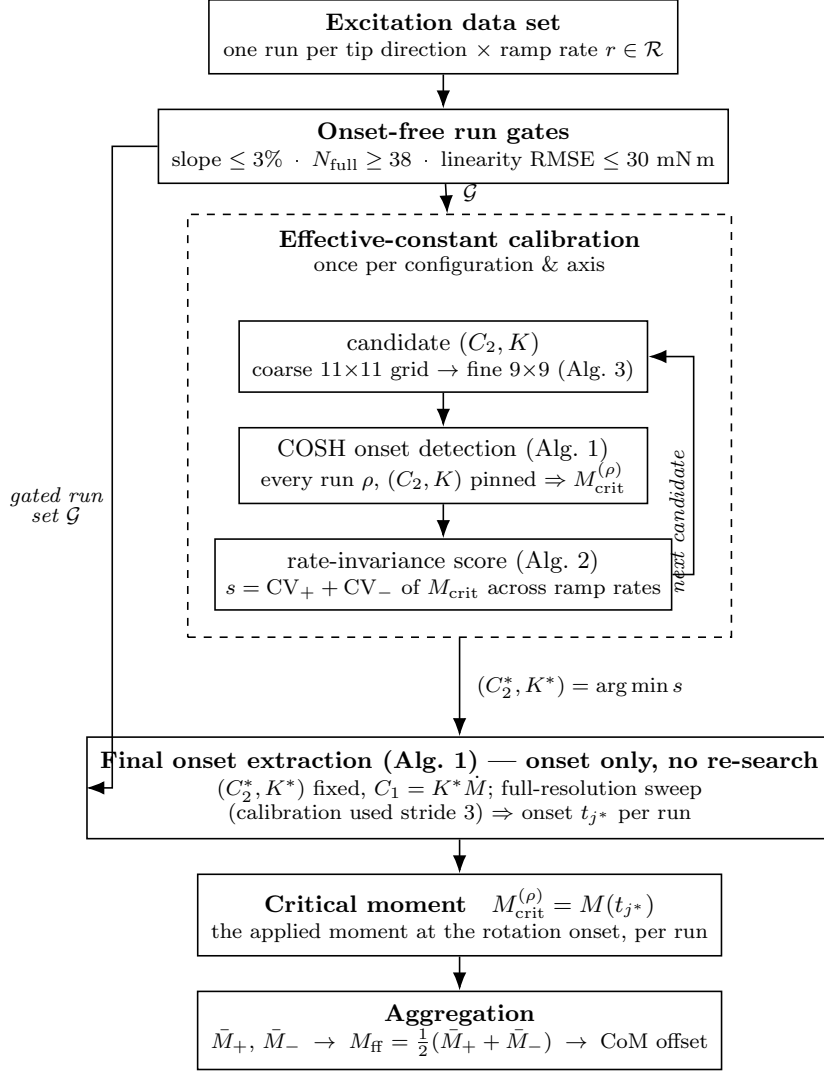


Figure 1: Pipeline at a glance. The dashed container is the once-per-configuration calibration: it iterates candidate pairs (C_2, K) , running the *same* onset detector (Algorithm 1) on every gated run and scoring the candidates by rate-invariance of the resulting critical moments. The selected pair then drives one final onset extraction per run — the constants are *not* re-estimated; only the onset index is re-swept, at full resolution, because the calibration subsamples the sweep by a stride of 3 for speed — whose onset sample yields $M_{\text{crit}} = M(t_{j^*})$, and the directional means aggregate into the pivot-free feed-forward moment.

5. **Aggregation** (Sec. 8) — directional means $\bar{M}_{\pm}$, the pivot-free feed-forward moment $M_{\text{ff}} = \frac{1}{2}(\bar{M}_+ + \bar{M}_-)$, and the CoM offset for validation against the load-cell truth (Sec. 9).

3 Contact-phase dynamics and the closed form

About the active pivot line, with the small-angle attitude dependence retained ($\cos \varphi \approx 1$, $\sin \varphi \approx \varphi$), the roll/pitch dynamics read

$$J_P \dot{\omega} = M(t) + f l_p - W(l_p + \lambda_{\text{off}}) + W z_{\text{CoM}} \varphi, \quad (1)$$

where M is the commanded body moment, f the collective thrust and λ_{off} the CoM offset along the excited axis. Below the critical value the vehicle is static. At the onset the net moment vanishes, which supplies *two* anchoring conditions,

$$\omega(t_{\text{crit}}) = 0, \quad \dot{\omega}(t_{\text{crit}}) = 0. \quad (2)$$

Past the onset gravity feeds back positively: the linearized dynamics have *real* eigenvalues $\pm C_2$ and, with the measured moment ramp $M(t) = M_{\text{crit}} + \dot{M} \tau$ in onset-relative time $\tau = t - t_{\text{crit}}$, the exact solution of

(1)–(2) collapses to the hyperbolic closed form

$$\omega(\tau) = C_1(\cosh(C_2\tau) - 1), \quad C_1 = K\dot{M}, \quad K = \frac{1}{Wz_{\text{CoM}}}, \quad C_2 = \sqrt{\frac{Wz_{\text{CoM}}}{J_P}} \quad (3)$$

for $\tau \geq 0$ (and $\omega = 0$ for $\tau < 0$). Because $J_P C_2^2 = Wz_{\text{CoM}}$, the amplitude C_1 is *fixed* by the measured ramp rate: with (C_2, K) pinned as per-configuration effective constants and the baseline fixed by continuity, **no shape parameter is free per run**. The time-quadratic model of the original submission is the $C_2 \rightarrow 0$ limit of (3) — it discards the gravity term that creates the instability, with 14–75% truncation error over the windows actually used — and is retained only as a baseline.

4 Signals and excitation window

Per run, the moment is reconstructed from the measured rotor speeds, $M(t) = C_T \sum_i b_{a,i} \Omega_i^2(t)$, and $\omega(t)$ is taken from the odometry rate. The window end is $i_{\text{end}} = \arg \max_t |M|$, *truncated at the first sample exceeding the allocator cap M_{cap}* : past it the thrust allocation saturates, the executed ramp is no longer linear, and the cosh family does not apply. (On the reference dataset the cap never binds; peak $|M| \leq 1.7$ N m.) The window start is the *last* sub-threshold sample before the peak, walking backwards — robust to spin-up blips that cross the threshold long before the actual ramp.

5 Onset-free run gates

Before any onset is estimated, each run must pass three gates evaluated on the measured moment trace alone (hence free of circularity):

1. **Ramp-slope error** $|\varepsilon| \leq 3\%$, with $\varepsilon = (|\dot{M}| - \dot{M}_{\text{cmd}})/\dot{M}_{\text{cmd}}$ and $\dot{M}$ the least-squares slope over the sub-segment $|M| \geq M_{\text{floor}}$ (0.7 N m roll, 0.4 N m pitch). The floors are the per-axis lower bounds of $|M_{\text{crit}}|$ over the ± 20 mm CoM-offset design box, so the gate scores the ramp exactly where an onset can occur: spin-up imperfections at near-zero moment cannot reject a run whose ramp is within spec where it matters. (On this dataset the floored gate passes all 140 runs; scored on the full window instead, 8 fast runs would be rejected by their early transients.) A 3% error in $\dot{M}$ perturbs $C_1 = K\dot{M}$ by 3% — about $7\times$ below the $\pm 20\%$ level the sensitivity analysis shows harmless.
2. **Window samples** $N_{\text{full}} \geq 38$: a necessary condition for any onset position. Because the window opens at $|M| = 10$ mN m while $M_{\text{crit}} \gtrsim 0.4$ N m, the onset cannot occur earlier than $\hat{N}_{\text{pre}} = M_{\text{crit}}^{\text{min}}/(\dot{M} T_s) \geq 30$ samples into the window even at the fastest ramp; adding the $N_{\text{seg}} = 8$ post-onset fit minimum gives 38.
3. **Linearity RMSE** ≤ 30 mN m about the run’s own best-fit line: a fault detector for aborted/stepped ramps, set at $10\times$ the execution noise floor and above the healthy-run maximum.

The gates run *before* calibration so a poorly executed ramp cannot contaminate the dataset-coupled constraint $C_1 = K\dot{M}$. Figure 2 and Table 2 report the three gate quantities for all 140 runs, evaluated on the floored segment: the slope tracking error stays within $\pm 2.4\%$ (median 0.28%), the ramp-linearity RMSE within 4.2 mN m (median 2.0 — the execution noise floor itself, a factor 7 below the gate), and the shortest window carries 53 samples ($1.4\times$ the necessary minimum).

6 Onset detection

With (C_2, K) pinned, detection reduces to an exhaustive sweep of the single remaining unknown — the onset index. Each candidate is a plain arithmetic evaluation: no optimizer, no initial guess, no seed model.

Two implementation details. *Baseline by median*. The pre-onset gyro trace is heavy-tailed (propeller vibration, spin-up spikes), and — more importantly — when a candidate j overshoots the true onset, the pre-segment tail contains the beginning of the hyperbolic rise. A mean baseline drifts with that leaked rise and partially cancels the residual it should expose, flattening the cost landscape; the median stays at the rest level until half the segment is contaminated, so misplaced candidates are penalized sharply and the minimum stays well defined. It is also a one-shot, seedless statistic, consistent with the no-optimizer

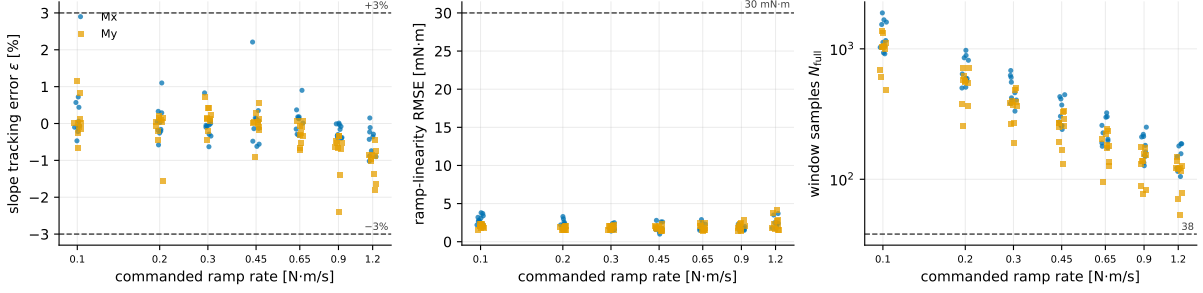


Figure 2: Run-gate quantities for all 140 runs versus the commanded ramp rate, in gate order: slope tracking error ε and ramp-linearity RMSE on the floored segment ($|M| \geq M_{\text{floor}}$), and the excitation-window sample count N_{full} . Dashed lines are the gate thresholds ($\pm 3\%$, 30 mN m, 38 samples); every run clears all three.

rate	$ \varepsilon $ [%]		lin. RMSE [mN m]		N_{full}	N_{floor}
[N m/s]	med	max	med	max	min	min
0.10	0.14	1.16	2.3	3.8	484	102
0.20	0.19	1.55	1.9	3.3	257	71
0.30	0.17	0.83	1.8	2.5	191	56
0.45	0.15	2.21	1.7	2.8	130	52
0.65	0.23	0.90	1.8	2.9	95	47
0.90	0.39	2.40	1.9	2.9	77	33
1.20	0.86	1.80	2.3	4.2	53	31

Table 2: Per-rate summary of the gate quantities (20 runs per rate; slope and linearity on the floored segment). All values are inside the gates with wide margin; the mild growth of $|\varepsilon|$ toward 1.20 N m/s reflects the shorter averaging segment, not degraded execution (`analysis/gate_report.py`).

design. An ablation rerunning the full pipeline — including the (C_2, K) calibration — with the mean instead (`analysis/baseline_stat_ablation.py`) leaves the run gating unchanged and 81/140 onsets identical, but biases the remainder toward *earlier* onsets, shrinking the directional means in magnitude by up to 30 mN m on the noisiest datasets. The route is indirect. At *fixed* (C_2, K) a baseline biased toward the motion direction *delays* the detected onset: the post-onset residual $C_1[\cosh(C_2(t - t_{\text{crit}}^{\text{real}})) - \cosh(C_2(t - t_{\text{crit}}^{\text{est}}))] - C$ can only vanish with a later split, and an oppositely biased baseline advances it. The dominant effect, however, is that the leaked-rise flattening of the cost landscape moves the CV-selected calibration to a different point on the (C_2, K) ridge — e.g. $C_2: 8.00 \rightarrow 5.38$, $K: 0.060 \rightarrow 0.150$ on the most affected dataset — whose net effect is the earlier-onset magnitude shrink; the ramp-invariance CV itself is essentially unchanged. The pivot-free offset M_{ff} moves by at most 4.0 mN m (0.13 mm), well below the run-to-run scatter — the median removes a systematic bias and tunes nothing. A second ablation pins $C = 0$, the ideal-sensor value implied by the onset conditions: the measured pre-onset baseline is in fact nonzero (gyro bias plus residual spin-up motion; $|C|$ median 3.5, p90 17.6, max 47.8 mrad/s), yet the estimate barely moves (66/140 onsets identical, M_{ff} shift ≤ 4.9 mN m, i.e. 0.16 mm). The estimated baseline is therefore not load-bearing on this dataset; it is kept because it makes the sweep invariant to the gyro bias — a vehicle- and temperature-dependent quantity that need not stay this small — at zero parameter cost. Nor does C need the bias to be constant in time: it is re-estimated per run, so only *within-window* drift (windows last 1–10 s) matters. Measured over the guaranteed pre-onset segment ($|M| < 0.3$ N m, spans ≤ 3.0 s), the baseline wanders by 3.4 mrad/s (median; p90 24, max 68) — an upper bound that conflates true sensor drift with real elastic pre-onset motion (`analysis/bias_drift.py`). This wander is present in the very data on which the $C = 0$ bound and the rate-invariance certificates were earned, and its structure is covered by the perturbation absorption: the constant part goes to C , a τ -linear trend is absorbed by the span projection, and only the residual curvature reaches ρ . *Sweep margin.* The margin $N_{\text{seg}} = 8$, applied at both window ends, guarantees every candidate split leaves at least N_{seg} samples in each segment; it excises only degenerate candidates: near $j = i_{\text{start}}$ the baseline would be estimated from a handful of samples, and near $j = i_{\text{end}}$ the post-onset arc is too short for the cosh branch to be falsifiable (at $j = i_{\text{end}}$ its cost is nearly an empty sum). Since physically feasible onsets lie at least $\hat{N}_{\text{pre}} \geq 30$ samples inside the window (Sec. 5) and the window closes only after the response is

Algorithm 1 COSH onset detection (single run)

Require: samples $\{(t_i, \omega_i)\}$ on the excitation window $[i_{\text{start}}, i_{\text{end}}]$; moment $M(t)$; calibrated (C_2, K) ; measured ramp slope $\dot{M}$; margin $N_{\text{seg}} = 8$

Ensure: onset index j^* ; critical moment M_{crit}

- 1: $C_1 \leftarrow K\dot{M}$ $\triangleright$ onset conditions (2) fix the amplitude
 - 2: **for** $j = i_{\text{start}} + N_{\text{seg}}, \dots, i_{\text{end}} - N_{\text{seg}}$ **do** $\triangleright$ exhaustive sweep of the single unknown
 - 3: $C \leftarrow \text{median}\{\omega_i : i_{\text{start}} \leq i < j\}$ $\triangleright$ baseline by continuity
 - 4: $\text{cost}(j) \leftarrow \sum_{i=i_{\text{start}}}^{j-1} (\omega_i - C)^2 + \sum_{i=j}^{i_{\text{end}}} (\omega_i - [C_1(\cosh C_2(t_i - t_j) - 1) + C])^2$
 - 5: **end for**
 - 6: $j^* \leftarrow \arg \min_j \text{cost}(j)$; $M_{\text{crit}} \leftarrow M(t_{j^*})$
-

well developed, the sweep remains exhaustive over every feasible onset.

Remark 1 *Because the model curve is fully determined before the post-onset data are seen, goodness of fit is a prediction test, not flexibility: across all gate-passing runs the normalized post-onset residual is $\approx 6\%$ (gyro-noise dominated) and invariant over the twelvefold ramp-rate range — the empirical certificate that the response stays in the cosh family.*

7 Calibration of the effective constants

Algorithm 1 presupposes (C_2, K) . Although both have geometric expressions (3), they are *not* computed from geometry: z_{CoM} and especially J_P are not precisely known a priori, and — more importantly — the constants that make (3) reproduce the measured response are *effective* values that absorb the state-linear contributions of the unmodeled effects (the local slope of the gravity nonlinearity and the attitude gradient of the ground effect). Both are calibrated once per dataset.

Joint residual minimization is not a usable criterion in practice. On the fitted window the amplitude and the onset are nearly interchangeable under noise: a smaller amplitude with an earlier onset and a larger amplitude with a later onset produce post-onset curves that differ most near the onset itself, where the response is quadratic, $\frac{1}{2}C_1C_2^2\tau^2$, and of the order of the gyro noise. In the noiseless limit the post-onset shape determines all three parameters uniquely — the trade-off is an ill-conditioning statement, not a non-identifiability one — but under gyro noise the summed residual is nearly flat along this amplitude–onset direction: a *shallow ridge* in the (C_2, K) plane, observed directly on the data (the objective is piecewise-constant in the integer onset index, gradient-based and simplex optimizers stall on it, and a population-based global search costs about $4\times$ more for no measurable gain). Per-run estimates — and with them the detected critical moments — scatter along the ridge. The calibration should therefore be read as *variance suppression*: choosing (C_2, K) once per dataset to minimize the dispersion of M_{crit} across ramp rates removes precisely that scatter, which is where the practical strength of the method lies.

The scatter direction is faint in the residual but loud in the deliverable: moving along the trade-off shifts each detected onset in *time*, and since $M_{\text{crit}} = M(t_{\text{on}})$ with M ramping at $\dot{M}$, an onset-time bias δt common across runs biases the critical moment by $\dot{M}\delta t$ — growing linearly with the ramp rate. A wrong pair thus *fans out* the identified M_{crit} across the twelvefold rate range, while the correct pair leaves it rate-invariant: exactly the direction a rate-consistency criterion sees best. We therefore calibrate by *physical self-consistency*: a static threshold must not depend on how fast it was approached. With runs at ramp rates $r \in \{0.10, \dots, 1.20\}$ N m/s,

$$\text{score}(C_2, K) = \sum_{\text{dir} \in \{+, -\}} \text{CV}\left(\{M_{\text{crit}}^{(r)}(C_2, K)\}_r\right), \quad \text{CV} = \text{std}/|\text{mean}|, \quad (4)$$

where $M_{\text{crit}}^{(r)}$ is the output of Algorithm 1 on the run with commanded rate r .

Search ranges. The admissible box is set so that it covers every physically plausible constant with margin. For $K = 1/(Wz_{\text{CoM}})$, the coarse range $K \in [0.10, 0.50]$ (N m) $^{-1}$ corresponds to $z_{\text{CoM}} \in [0.06, 0.32]$ m at the vehicle weight — bracketing the plausible CoM heights — with the refinement allowed to move within the wider clip box $[0.05, 0.70]$ (z_{CoM} up to 0.63 m). For $C_2 \in [3, 8]$ rad/s, the implied pivot inertia $J_P = 1/(KC_2^2)$ spans roughly $[0.03, 1.1]$ kg m 2 , bracketing the parallel-axis estimate $J + mz_{\text{CoM}}^2 \approx 0.2$ – 0.35 kg m 2 by a factor of several on both sides.

Algorithm 2 Ramp-invariance consistency score

Require: set $\mathcal{G}$ of gated runs (both tip directions, spanning the rate set $\{0.10, \dots, 1.20\}$ N m/s); candidate pair (C_2, K)
Ensure: consistency score s $\triangleright$ lower = M_{crit} more rate-invariant

- 1: **for** $\text{dir} \in \{+, -\}$ **do**
- 2: $\mathcal{G}_{\text{dir}} \leftarrow \{\rho \in \mathcal{G} : \text{dir}(\rho) = \text{dir}\}$
- 3: **for** each run $\rho \in \mathcal{G}_{\text{dir}}$ **do**
- 4: $M_{\text{crit}}^{(\rho)} \leftarrow$ Algorithm 1 with (C_2, K) pinned $\triangleright$ stride-3 sweep
- 5: **end for**
- 6: $\text{CV}_{\text{dir}} \leftarrow \text{std}(\{M_{\text{crit}}^{(\rho)}\}_{\rho \in \mathcal{G}_{\text{dir}}}) / |\text{mean}(\{M_{\text{crit}}^{(\rho)}\}_{\rho \in \mathcal{G}_{\text{dir}}})|$
- 7: **end for**
- 8: **return** $s \leftarrow \text{CV}_+ + \text{CV}_-$ $\triangleright$ Eq. (4): a static threshold must not depend on the approach rate

Note that the ramp rate never appears explicitly in the score: each direction’s gated runs span the full rate set, so the within-direction dispersion of the per-run critical moments *is* the rate dependence (plus repeatability scatter, which the score harmlessly co-penalizes). Should a rate be repeated, each run enters individually.

Algorithm 3 Effective-constant calibration (coarse-to-fine)

Require: gated runs; box $C_2 \in [3, 8]$ rad/s, $K \in [0.05, 0.70]$ (N m) $^{-1}$
Ensure: calibrated pair (C_2^*, K^*)

- 1: **coarse:** evaluate Algorithm 2 on the 11×11 grid $C_2 = 3.0:0.5:8.0$, $K = 0.10:0.04:0.50$
- 2: $(C_2^b, K^b) \leftarrow$ arg min over the coarse grid
- 3: **fine:** evaluate Algorithm 2 on a 9×9 uniform grid over $[C_2^b \pm 0.5] \times [K^b \pm 0.04]$, clipped to the box $\triangleright$ resolution 0.125 rad/s, 0.01 (N m) $^{-1}$
- 4: $(C_2^*, K^*) \leftarrow$ arg min over the fine grid

Coarse-to-fine procedure. Algorithm 2 states the consistency criterion as a stand-alone routine — it wraps Algorithm 1 and returns the directional sum of coefficients of variation — and Algorithm 3 performs the search over it. The coarse scan evaluates (4) on an 11×11 grid ($\Delta C_2 = 0.5$ rad/s, $\Delta K = 0.04$ (N m) $^{-1}$; 121 evaluations). A single refinement pass then re-scans a 9×9 uniform grid spanning $\pm$ one coarse step about the coarse minimizer (± 0.5 rad/s, ± 0.04 (N m) $^{-1}$, clipped to the box), i.e. final resolutions of 0.125 rad/s and 0.01 (N m) $^{-1}$ — so the reported constants are not quantized to the coarse step. One pass suffices because the objective is flat along the (C_2, K) ridge; it is also *piecewise constant* (the onset is an integer index, so small parameter changes leave M_{crit} unchanged), which is why gradient-based or simplex optimizers stall on it, and a population-based global search (differential evolution) was found to cost about $4 \times$ more for no measurable gain. During calibration the onset sweep of Algorithm 1 is subsampled by a stride of 3 samples for speed; the final identification runs at full resolution with the selected pair. Each evaluation is an arithmetic sweep, so no iterative optimizer is involved anywhere in the pipeline.

Remark 2 *The objective (4) retains a shallow valley, so the individual constants are only loosely determined: $W z_{\text{CoM}} = 1/K$ and $J_P = 1/(K C_2^2)$ are order-of-magnitude sanity checks, not precision measurements. What the criterion pins down — by construction — is the identified onset: M_{crit} is robust along the valley, and a $\pm 20\%$ mis-specification of either constant moves the identified CoM by less than 0.9 mm.*

8 Aggregation and deliverables

Directional means over the gated runs give $\bar{M}_+$ and $\bar{M}_-$. The take-off feed-forward moment is the *pivot-free average*

$$M_{\text{ff}} = \frac{1}{2} (\bar{M}_+ + \bar{M}_-), \quad (5)$$

in which the pivot-arm terms $\pm(W - f)l_p$ — and any thrust-proportional contamination acting through the arm l_p , ground effect included — cancel by antisymmetry, while landing-gear asymmetry is deliberately *retained*: the same contact-phase moment acts at lift-off, where M_{ff} compensates it. The CoM

case	axis	λ_{truth} [mm]	estimated CoM offset $\hat{\lambda} \pm \text{CI}_{95}$ [mm]				
			COSH	NLS	CPD _N	CPD _R	CUSUM
1	x	-2.90	-1.73 $\pm$ 0.93	-2.05 $\pm$ 1.30	-1.46 $\pm$ 1.03	-1.20 $\pm$ 1.51	-1.15 $\pm$ 1.24
1	y	-11.45	-11.13 $\pm$ 0.43	-11.35 $\pm$ 1.37	-11.21 $\pm$ 0.61	-10.79 $\pm$ 0.80	-9.59 $\pm$ 3.88
2	x	-14.29	-15.17 $\pm$ 0.36	-15.03 $\pm$ 0.76	-15.09 $\pm$ 0.70	-14.10 $\pm$ 4.15	-15.00 $\pm$ 0.52
2	y	-9.90	-10.49 $\pm$ 0.42	-9.60 $\pm$ 1.13	-9.91 $\pm$ 0.52	-9.61 $\pm$ 0.46	-8.54 $\pm$ 4.02
3	x	-5.26	-8.62 $\pm$ 0.46	-8.54 $\pm$ 0.74	-8.11 $\pm$ 1.20	-4.63 $\pm$ 4.72	-8.45 $\pm$ 0.92
3	y	+3.14	+0.73 $\pm$ 0.39	+1.06 $\pm$ 0.97	+0.71 $\pm$ 0.82	+0.94 $\pm$ 1.00	+1.12 $\pm$ 3.39
4	x	+6.67	+5.26 $\pm$ 0.65	+4.94 $\pm$ 0.88	+5.57 $\pm$ 0.72	+3.58 $\pm$ 2.19	+5.01 $\pm$ 0.91
4	y	+2.40	+0.16 $\pm$ 0.61	-0.64 $\pm$ 1.49	-0.66 $\pm$ 1.39	-0.60 $\pm$ 1.36	-0.22 $\pm$ 1.36
5	x	+10.91	+10.70 $\pm$ 0.69	+10.05 $\pm$ 1.10	+10.29 $\pm$ 0.65	+11.16 $\pm$ 3.75	+10.25 $\pm$ 0.74
5	y	-10.89	-10.73 $\pm$ 0.70	-11.11 $\pm$ 1.68	-10.84 $\pm$ 0.98	-10.95 $\pm$ 0.95	-10.84 $\pm$ 1.00
median err [mm]			1.02	0.86	0.95	0.64	1.70
RMS err [mm]			1.64	1.72	1.67	1.65	1.82
max err [mm]			3.36	3.28	3.06	3.09	3.19

Table 3: CoM offset from the pivot-free average (5) against the load-cell ground truth (manuscript Table 7; per-case weight 3.066 kg for case 1, 3.220 kg otherwise), for the five onset estimators on identical inputs. $M_{\text{ff}} = \pm W \hat{\lambda}$ per the axis mapping of Sec. 8.

offset follows from the moment-to-offset conversion of M_{ff} and is used for validation against load-cell ground truth only; M_{ff} itself is the quantity applied in flight. A roll excitation (M_x) senses the y -offset via $M_{\text{ff},x} = +W y_{\text{off}}$; a pitch excitation (M_y) senses the x -offset via $M_{\text{ff},y} = -W x_{\text{off}}$.

9 Comparison with per-run estimators: NLS, CPD, CUSUM

Four per-run alternatives were run on *identical* inputs — same signals, excitation windows, allocator caps and onset-free gates (the gated run set is method-independent because the gates use the moment trace only), 140 runs in total (analysis/nls_comparison.py):

- **NLS** — the predecessor of Algorithm 1: per-run nonlinear least squares (SCIPY `least_squares`, trust-region reflective), free (C_1, C_2, C) with C_2 bounded to $[3, 8]$ rad/s, onset swept locally around a quadratic-model seed;
- **CPD_N / CPD_R** — optimal single change-point detection (`ruptures`; with one break the search over all admissible splits is exact) under the Gaussian mean+variance cost and the nonparametric RBF-kernel cost, respectively;
- **CUSUM** — the tabular sequential detector with allowance $k = 2\sigma$ against the pre-onset vibration.

Figure 3 is the readable summary: per case-axis pair it shows each estimator’s CoM offset with its 95% confidence interval (Welch t on the pivot-free average, run-to-run scatter) against the load-cell truth. The full data sit in the tables: Table 3 compares the deliverable — the CoM offset from the pivot-free average (5) — against the load-cell ground truth; Table 4 lists the underlying directional critical-moment means, pivot-free offsets and their confidence intervals; and Table 5 compares the ramp-rate dispersion of the directional critical moments.

Three observations. (i) *The deliverable is detector-agnostic*: against the load-cell truth all five estimators land at RMS 1.6–1.8 mm (Table 3, bottom rows) — the pivot-free average (5) cancels each method’s largely common-mode onset bias between the + and – directions, and 5–7 runs per direction average the rest. That the final CoM offset does not depend on the detection methodology is itself a validation certificate — and Fig. 4 sharpens it: the residual errors that do remain (cases 3–4) are shared by all five estimators, placing them in the vehicle–contact geometry rather than in any onset detector. (ii) *Run-level robustness is where the methods separate*: COSH has the lowest CV in 16/20 direction groups (CPD_N takes 3, CPD_R 1), median 2.5% against 4.1–7.1%, and its worst group is 5.7% where the alternatives reach 8.6–27.4% — the nonparametric detectors (CPD_R, CUSUM) occasionally misplace the onset badly on short fast-ramp windows. Equivalently, the 95% confidence interval on the CoM offset is ± 0.3 –0.9 mm under COSH — the narrowest in 9 of 10 case-axis pairs — against up to ± 4 –5 mm for the nonparametric detectors (Fig. 3). The paired per-run critical moments (NLS vs. COSH) differ by 33 mNm at the median but up to 269 mNm — the ridge of Sec. 7 realized run by run. (iii) *The biases carry signatures*: NLS sits 3.7% low in mean $|M|$ (the early-onset drift of the free amplitude–onset fit),

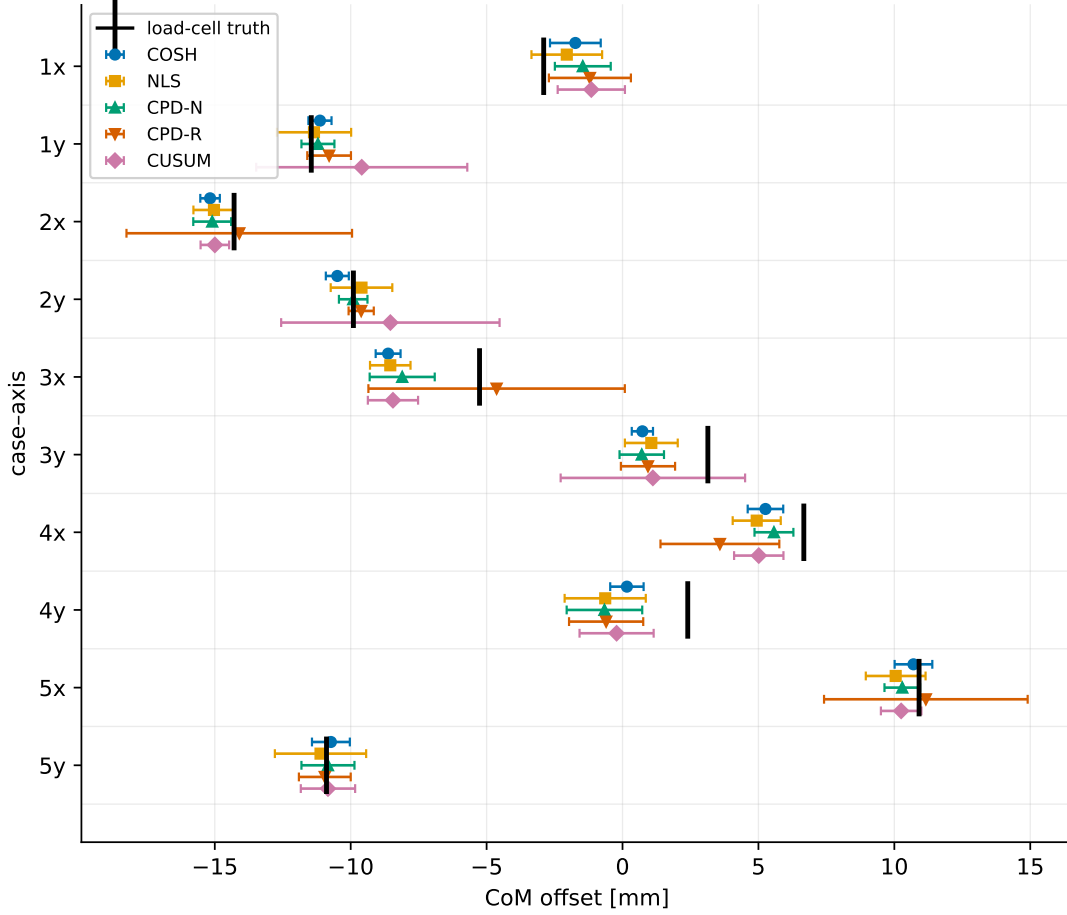


Figure 3: Estimated CoM offset with 95% confidence intervals for the five estimators, against the load-cell truth (black tick), per case-axis pair. COSH (blue) has the narrowest interval in 9 of 10 pairs (± 0.3 – 0.9 mm); the nonparametric detectors blow up to ± 4 – 5 mm on the pairs where they misplace onsets (CPD_R at 2x, 3x, 5x; CUSUM at 1y, 2y, 3y). Marker shapes duplicate the color coding.

while CPD_N and CUSUM sit 3.0% and 3.1% *high* — detection lag, since a distribution change must accumulate post-onset evidence before it is declared; both signatures largely cancel in (5). COSH needs no per-run tuning (CUSUM’s allowance k , CPD’s minimum segment) and a *single* run already sits within a few percent of its directional mean — which is what enables run-count reduction and per-run gating diagnostics.

10 Per-direction check against the static thresholds

The static tip-over thresholds tie each directional critical moment to independently measured quantities: $M_{\pm} = \pm(W - f)l_{\text{arm}} + Wy_{\text{off}}$ (roll) and $M_{\pm} = \pm(W - f)l_{\text{arm}} - Wx_{\text{off}}$ (pitch), with W and the offsets pinned by the load-cell truth, f the measured collective thrust at the onset, and the pivot arms fitted from the *position* trace: a circle fit of the odometry motion about the contact line per run (`estimate_pivot_from_mocap`); each run is predicted with its own arm and onset thrust, and the per-run predictions and residuals are then averaged per case/axis/direction. The kinematic arms come out remarkably consistent — 140.4 ± 2.3 mm (roll) and 112.7 ± 6.6 mm (pitch) across the five configurations, matching the geometric values (140/110 mm). Every ingredient is independent of the identified moments, so this is a genuine forward test — and no simulation enters anywhere: measured trajectories are combined with closed-form coefficients evaluated at the measured geometry.

Ground effect enters through *both* channels of the pivot-moment decomposition $\Delta M_{\text{GE}} = \text{sgn} \cdot a + bM$ (`analysis/ge_linearity.py`): the thrust channel $a = c_a f l$ (rotor-common surplus plus the rotor-differential moment, arm exactly l) and the moment-proportional channel b . The onset balance $M(1 +$

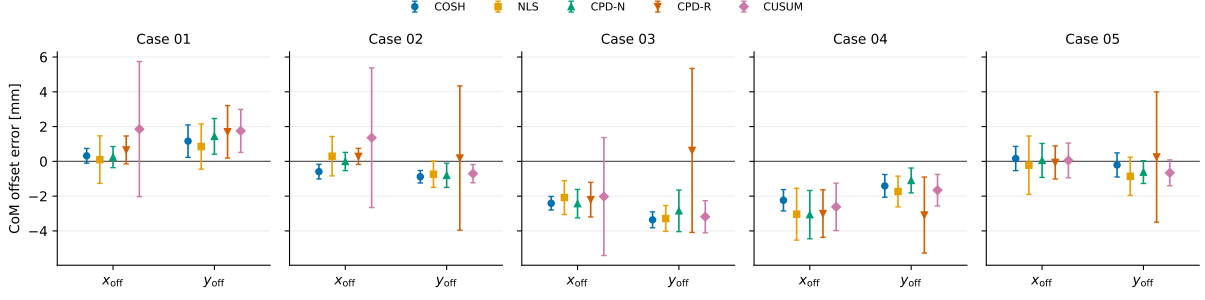


Figure 4: The same data folded about the truth: CoM offset *error* (estimate – truth) with 95% confidence intervals, grouped by component within each case (x_{off} from the pitch excitation, y_{off} from the roll excitation). The residual biases of cases 3 and 4 are *common to all five estimators* — a vehicle–contact asymmetry, not a detection artifact — while the estimator-to-estimator differences stay within about ± 1 mm.

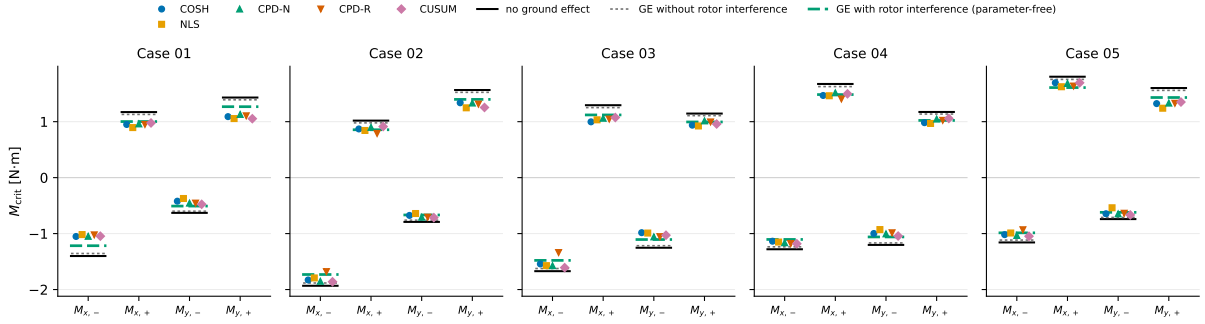


Figure 5: Identified directional critical moments (five estimators, colored markers) against the static-threshold predictions with truth offsets and odometry-fitted arms: no ground effect (black solid), ground effect without rotor interference (gray dotted), and with rotor interference (green dashed). The interference prediction passes through the estimator cluster in nearly every group, while the other two sit systematically outside it.

$b) + \text{sgn } a = T_{\text{rigid}}$ gives

$$M_{\text{pred}} = \frac{T_{\text{rigid}} - \text{sgn} \cdot c_a f l}{1 + b}, \quad T_{\text{rigid}} = \text{sgn}(W - f) l + S_{\text{off}}, \quad (6)$$

evaluated at three levels of physics: (a) no ground effect; (b) ground effect *without* rotor–rotor interference — the per-rotor Cheeseman–Bennett superposition at the tilted heights ($c_a = 1.03\%$, $b = 1.03\%$); and (c) ground effect *with* rotor interference — the image superposition over the rotor-centre distances, $s_{\text{rot},j} = (R^2/4) \sum_n (z_j + z_n) / [d_{jn}^2 + (z_j + z_n)^2]^{3/2}$, $\gamma_j = (1 - s_{\text{rot},j})^{-1}$, whose self term reduces exactly to Cheeseman–Bennett and which contains *no empirical constant of any kind* ($c_a = b = 4.31\%$).

Three observations. (i) *Ground effect is required, and rotor interference with it.* Without ground effect the prediction over-predicts the threshold magnitude in all 20 direction groups by 199 mNm on average — a strictly one-sided gap, far above the identification scatter and shared by all five estimators (last column), hence physical rather than algorithmic. Adding the single-rotor ground effect removes only a fifth of it (–159 mNm, still 20/20 one-sided): neglecting rotor–rotor interference is not tenable at this clearance. (ii) *The parameter-free interference model closes the gap.* With interference included, the median residual falls to 61 mNm (RMS 86), the mean magnitude deficit to –36 mNm, and the one-sidedness breaks (12/20) — a model with *no* tuned constant, evaluated at the measured geometry and driven by measured thrust, reproduces twenty independent directional thresholds at the $\sim 5\%$ level. That is a forward validation of the ground-effect treatment, not a fit. (iii) *What remains is contact physics, and it cancels.* The residual –36 mNm mean deficit is consistent with contact-patch compliance — rotation onsets when the normal-force resultant reaches the *inner* edge of a finite, compliant contact patch, slightly before the rigid-edge threshold — which is degenerate with the *a*-channel at this level and cannot be separated here. All these terms enter the two directions antisymmetrically through $\pm(W - f)l$

method		1x	1y	2x	2y	3x	3y	4x	4y	5x	5y
COSH	$\bar{M}_-$	-1.054	-0.421	-1.829	-0.675	-1.542	-0.983	-1.136	-0.998	-1.019	-0.645
	$\bar{M}_+$	+0.950	+1.090	+0.870	+1.338	+0.997	+0.937	+1.468	+0.988	+1.695	+1.323
	M_{ff}	-0.052	+0.335	-0.479	+0.331	-0.272	-0.023	+0.166	-0.005	+0.338	+0.339
	CI_{95}	± 0.028	± 0.013	± 0.011	± 0.013	± 0.014	± 0.012	± 0.021	± 0.019	± 0.022	± 0.022
NLS	$\bar{M}_-$	-1.018	-0.373	-1.792	-0.641	-1.573	-0.990	-1.151	-0.928	-0.989	-0.539
	$\bar{M}_+$	+0.894	+1.056	+0.843	+1.248	+1.033	+0.923	+1.463	+0.969	+1.624	+1.241
	M_{ff}	-0.062	+0.341	-0.475	+0.303	-0.270	-0.033	+0.156	+0.020	+0.317	+0.351
	CI_{95}	± 0.039	± 0.041	± 0.024	± 0.036	± 0.024	± 0.031	± 0.028	± 0.047	± 0.035	± 0.053
CPD_N	$\bar{M}_-$	-1.048	-0.456	-1.852	-0.707	-1.573	-1.059	-1.164	-1.005	-1.033	-0.647
	$\bar{M}_+$	+0.960	+1.130	+0.898	+1.333	+1.061	+1.014	+1.516	+1.047	+1.683	+1.332
	M_{ff}	-0.044	+0.337	-0.477	+0.313	-0.256	-0.022	+0.176	+0.021	+0.325	+0.342
	CI_{95}	± 0.031	± 0.018	± 0.022	± 0.017	± 0.038	± 0.026	± 0.023	± 0.044	± 0.021	± 0.031
CPD_R	$\bar{M}_-$	-1.023	-0.455	-1.680	-0.706	-1.339	-1.055	-1.179	-0.986	-0.932	-0.637
	$\bar{M}_+$	+0.950	+1.105	+0.789	+1.314	+1.046	+0.995	+1.406	+1.024	+1.637	+1.328
	M_{ff}	-0.036	+0.325	-0.445	+0.304	-0.146	-0.030	+0.113	+0.019	+0.353	+0.346
	CI_{95}	± 0.045	± 0.024	± 0.131	± 0.015	± 0.149	± 0.032	± 0.069	± 0.043	± 0.118	± 0.030
CUSUM	$\bar{M}_-$	-1.046	-0.476	-1.861	-0.714	-1.608	-1.029	-1.181	-1.044	-1.048	-0.667
	$\bar{M}_+$	+0.977	+1.054	+0.913	+1.254	+1.075	+0.958	+1.498	+1.058	+1.695	+1.351
	M_{ff}	-0.035	+0.289	-0.474	+0.270	-0.267	-0.035	+0.158	+0.007	+0.324	+0.342
	CI_{95}	± 0.037	± 0.117	± 0.017	± 0.127	± 0.029	± 0.107	± 0.029	± 0.043	± 0.024	± 0.032

Table 4: Directional critical-moment means $\bar{M}_\pm$, pivot-free offset M_{ff} and its 95% confidence interval (Welch t) [Nm] per estimator. Columns are case-axis pairs (e.g. 1x = case 1, roll). Note the signed signatures relative to COSH: NLS magnitudes sit low (early onsets), CPD_N /CUSUM high (detection lag), while M_{ff} stays nearly method-invariant.

and $\pm a$, so they cancel in the pivot-free deliverable (5): M_{ff} validates to millimetres (Sec. 9) while the individual thresholds carry a ~ 0.06 N m ground-contact budget.

11 Reproducibility

Every number in this document is reproducible from the repository:

- `critical_value_getter_piecewise.py` — pipeline, gates, caps, calibration, Algorithm 1;
- `analysis/ramp_quality.py` — gates;
- `analysis/sensitivity.py` — $\pm 20\%$ robustness;
- `analysis/cosh_fidelity.py` — prediction-test residuals;
- `analysis/excitation_angle_design.py` — rate/cutoff design;
- `analysis/ge_linearity.py`, `analysis/ge_trajectory.py` — ground-effect bracket;
- `analysis/tiltcap_ablation.py` — window ablation;
- `analysis/baseline_stat_ablation.py` — median/mean/zero baseline;
- `analysis/bias_drift.py` — in-window baseline drift;
- `analysis/nls_comparison.py` — Tables 3–5;
- `analysis/mcrit_prediction.py`, `analysis/mcrit_static_figure.py` — Sec. 10 (static-threshold check, arms, Fig. 5).

case	axis	COSH		NLS		CPD _N		CPD _R		CUSUM	
		CV ₋	CV ₊	CV ₋	CV ₊	CV ₋	CV ₊	CV ₋	CV ₊	CV ₋	CV ₊
1	x	3.8	5.7	6.4	7.7	4.4	6.1	7.3	8.6	5.9	6.7
1	y	4.2	2.3	17.1	7.2	5.0	3.3	7.0	4.4	4.8	24.0
2	x	1.2	1.6	2.5	4.2	2.5	2.2	15.5	22.1	1.8	2.4
2	y	1.5	2.1	7.1	5.8	3.1	2.5	3.3	2.0	3.0	21.9
3	x	1.7	2.4	2.4	4.1	4.9	4.2	23.9	8.5	3.1	4.7
3	y	1.7	2.6	3.2	7.0	4.4	4.1	4.4	6.0	12.4	22.8
4	x	3.4	2.2	3.8	3.5	4.0	1.6	5.6	10.3	3.9	3.5
4	y	2.9	3.7	10.3	6.1	7.5	7.2	7.8	6.9	6.7	7.3
5	x	3.9	2.1	7.2	2.6	3.6	2.0	27.4	3.0	4.0	2.3
5	y	5.3	3.1	19.8	5.4	8.6	3.8	8.4	3.8	8.3	4.0
median / worst [%]		2.5 / 5.7		5.9 / 19.8		4.1 / 8.6		7.1 / 27.4		4.7 / 24.0	

Table 5: Ramp-rate dispersion (CV [%]) of the directional critical moments under the five estimators. Bold marks the lowest CV in each of the 20 direction groups: COSH in 16, CPD_N in 3, CPD_R in 1, the rest in none. The underlying directional means are in Table 4.

thrust model	residual [mN m]				magnitude deficit	
	median	p90	max	RMS	mean [mN m]	one-sided
(a) no ground effect	204.8	301.5	349.5	213.3	-199.4	20/20
(b) GE, no rotor interference	163.9	258.9	304.2	176.1	-159.1	20/20
(c) GE with rotor interference	61.0	127.6	177.4	86.0	-35.8	12/20

Table 6: Residuals of the identified directional means against the static thresholds (odometry-fitted arms held fixed). The magnitude deficit is $|M_{\text{ident}}| - |M_{\text{pred}}|$, whose sign exposes systematic over-prediction (the raw signed residual hides it, the two tip directions carrying opposite signs); one-sided” counts the groups over-predicted. Each added physics element removes part of a strictly one-sided bias, and the parameter-free interference model leaves no one-sidedness worth the name.

case	axis	dir	l_{odom} [mm]	COSH $\bar{M}$ [N m]	no GE		no int. resid	with interference		benchmark range [N m]
					pred	resid		pred	resid	
1	x	-	145.2	-1.051	-1.401	+350	+304	-1.217	+165	[-1.048, -1.018]
1	x	+	139.2	+0.950	+1.172	-222	-180	+1.002	-52	[+0.894, +0.977]
1	y	-	109.3	-0.421	-0.630	+209	+179	-0.508	+87	[-0.476, -0.373]
1	y	+	120.1	+1.090	+1.431	-341	-301	+1.268	-177	[+1.054, +1.130]
2	x	-	140.0	-1.829	-1.933	+104	+54	-1.731	-98	[-1.861, -1.680]
2	x	+	139.4	+0.870	+1.019	-149	-109	+0.856	+14	[+0.789, +0.913]
2	y	-	105.0	-0.675	-0.791	+116	+86	-0.667	-8	[-0.714, -0.641]
2	y	+	118.5	+1.338	+1.566	-228	-187	+1.398	-60	[+1.248, +1.333]
3	x	-	142.5	-1.542	-1.672	+130	+82	-1.479	-63	[-1.608, -1.339]
3	x	+	138.3	+0.997	+1.294	-297	-254	+1.121	-123	[+1.033, +1.075]
3	y	-	109.3	-0.984	-1.253	+270	+233	-1.106	+123	[-1.059, -0.990]
3	y	+	117.9	+0.937	+1.145	-208	-171	+0.995	-58	[+0.923, +1.014]
4	x	-	141.1	-1.136	-1.280	+144	+101	-1.104	-32	[-1.181, -1.151]
4	x	+	138.4	+1.468	+1.674	-206	-159	+1.484	-16	[+1.406, +1.516]
4	y	-	106.7	-0.998	-1.202	+204	+169	-1.060	+62	[-1.044, -0.928]
4	y	+	118.3	+0.982	+1.174	-192	-154	+1.022	-40	[+0.969, +1.058]
5	x	-	142.5	-1.019	-1.158	+140	+97	-0.987	-32	[-1.048, -0.932]
5	x	+	137.9	+1.695	+1.804	-109	-61	+1.609	+86	[+1.624, +1.695]
5	y	-	103.2	-0.645	-0.741	+96	+66	-0.620	-25	[-0.667, -0.539]
5	y	+	118.9	+1.323	+1.601	-278	-236	+1.432	-109	[+1.241, +1.351]

Table 7: Per-direction static-threshold check with odometry-fitted arms; residuals in mN m. The last column pair is the parameter-free rotor-interference model; the rightmost entry of each row is the range spanned by the four alternative estimators (NLS, CPD_N, CPD_R, CUSUM), showing that the disagreement with the static prediction is not an artifact of the onset detector.